John Joseph Youkhana

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Education

University of Illinois Urbana-Champaign — Urbana, IL

Bachelor of Science in Electrical Engineering — *Aug 2024 – May 2027*

- **Relevant Coursework:** Power Electronics & Laboratory (ECE 464/469), Control Systems (ECE 486), Wireless Communication Systems (ECE 453), Digital IC Design (ECE 482), Analog IC Design (ECE 483)

Projects

FM Transmitter — Nov 2027

- Designed a 2-layer FM transmitter PCB in KiCad around the MAX2606 VCO (88–108 MHz band), integrating adjustable carrier tuning, electret microphone voice modulation, a 3.3 V LDO supply, and an SMA RF test port.
- Applied EMI-conscious layout: ground planes on both layers with stitching vias, ground-plane keep-outs under the VCO tank and SMA signal pad to reduce parasitic capacitance, and local supply decoupling.
- Validated prototype RF performance on a spectrum analyzer, measuring a 97.5–105.5 MHz carrier tuning range at -18 to -16 dBm output power with voice-modulated FM sidebands

Synchronous Buck Converter with Feedback Control — In Progress

- Designing a synchronous buck converter regulating 10 V within $\pm 5\%$ across a 10-20 W (1-2 A) load range from a 12.5-33 V input, with duty cycle bounded to 30-80%.
- Designing a closed-loop voltage controller with ADC feedback, PI compensation, and input-voltage feedforward, and verifying stability and transient response at the input and load extremes.
- Generating complementary PWM from a micro-controller timer to drive a half-bridge gate driver, with dead time to prevent shoot-through between the high- and low-side MOSFETs.

Reaction Wheel Pendulum Control System — May 2026

- Designed and implemented an observer-based state-space controller for a reaction wheel pendulum, achieving swing-up and inverted stabilization via DC motor torque.
- Linearized the plant about the upright equilibrium and designed full-state feedback by pole placement, with a state observer estimating the unmeasured velocity states.
- Conducted experimental system identification and PI speed-control tests to model plant parameters and implement real-time static and dynamic motor friction compensation.

Low-Dropout Regulator — May 2026

- Designed a CMOS LDO in Cadence Virtuoso (180nm) delivering 1.2 V across a 0.1–10 mA load range using a folded-cascode error amplifier with reference voltage and feedback.
- Optimized pass-transistor sizing and feedback network values, implementing pole-splitting compensation via gate-drain capacitance across the pass MOSFET.
- Achieved $< 14\mu$V/mA DC load regulation and $< 60\mu$V/V DC line regulation at 1mA, loop-gain phase margin of 66.2° to 87° across full load range while minimizing quiescent current.

Experience

Electrical Team Member — Aug 2024 – Present

Illini Solar @ UIUC — *Urbana, IL*

- Designed PCB layouts in KiCad for a battery charge switch and horn oscillator, including component placement, routing, and DRC verification; assisted with board assembly of SMD and THT devices via soldering.
- Verified assembled boards through continuity and power checks to confirm functionality prior to integration.

Engineering Tutor — Aug 2023 – May 2024

City Colleges of Chicago at Wilbur Wright College — *Chicago, IL*

- Tutored students 1-on-1 and in small groups across Calculus I–III, Physics I–II, Circuit Analysis, and C++.

Technical Skills

Instrumentation: Power Meter, Digital Oscilloscope, Spectrum Analyzer, Vector Network Analyzer (VNA), RF Signal Generator, 3-Phase AC Power

Software & CAD: MATLAB, Simulink (Simscape, Stateflow), Cadence Virtuoso, Keysight ADS, LTspice, HSPICE, KiCad, Autodesk Fusion

Programming: C/C++, Python (NumPy, SciPy), SystemVerilog, VHDL, Git